# Zijian Lu

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## Education

2024–Present **Nanjing University of Posts and Telecommunications**
B.Eng. candidate in Communication Engineering — Expected 2028
School of Communication and Information Engineering

## Research Interests

Reliable multimodal agents; speech and language technologies for cognitive screening; contactless sensing and evidence-grounded health AI.

## Selected Publications & Challenge Paper

### Conference papers

[1] **ContractSkill: Repairable Contract-Based Skills for Multimodal Web Agents**.
**Zijian Lu**, Yiping Zuo, Yupeng Nie, Xin He, Weibei Fan, Lianyong Qi, Shi Jin.
*ACM International Conference on Multimedia (ACM MM)*, 2026 **[Accepted; Poster]** [DOI] [Code].

[2] **Why Alzheimer's Speech Screening Fails to Generalize: Bridging the Deployment Gap via Cross-Corpus Evidence Anchoring**.
**Zijian Lu**, Sizhe Liu, Yin Zhang, Jixuan Deng, Xinrong Lin, Xinchen Yuan, Chicheng Jin, Yiping Zuo, Yuanchao Li.
*NCMMSC*, 2026 **[Accepted; Oral]**.

### Student challenge

[3] **SuperSenseDoctor: A Multimodal and Contactless Agent for Health Tracking**.
Xuwen Zhang, **Zijian Lu**, Yicheng Lei, Rui Qiu, Jiale Li, Yiping Zuo, Weibei Fan, Fu Xiao.
*UbiComp/ISWC Student Challenge*, 2026 **[Accepted; Student Challenge]**.

## Research Experience

**ContractSkill: Repairable Skills for Multimodal Web Agents** 2026
*First author | ACM MM 2026, Poster*

- Developed a framework that compiles model-generated skills into executable contracts with explicit preconditions, step constraints, postconditions, and recovery rules; combined deterministic verification with step-level fault localization and minimal patch repair.
- Evaluated GLM-4.6V and Qwen3.5-Plus on VisualWebArena and MiniWoB. On the evaluated MiniWoB split, Qwen success increased from 60.5% with unverified self-generated skills to 81.0%; tested cross-model reuse under matched transfer settings.

**Cross-Corpus Speech Screening for Cognitive Risk** 2026
*First author | NCMMSC 2026, Oral*

- Audited 70 speech and language markers across four English and Mandarin corpora under leave-one-corpus-out evaluation; identified cross-corpus direction conflicts in 59 markers.
- Developed training-only evidence-anchor selection and score fusion with XLM-R. Balanced fusion achieved a mean speaker AUC of 0.785; anchor-heavy fusion raised worst-domain AUC from 0.520 to 0.615.

## Research Experience (continued)

**SuperSenseDoctor: Contactless Health Tracking** 2026
*Second author | UbiComp/ISWC 2026 Student Challenge*

- Developed contactless sensing and multimodal fusion algorithms combining WiFi beamforming feedback and mmWave radar with thermal context for continuous physiological monitoring.
- Designed confidence-aware arbitration for heart-rate and respiratory-rate estimation under sensing-quality changes and modality conflicts; retained quality indicators and provenance in a shared health-state representation.
- Evaluated the prototype on 2,686 one-second states across nine recording intervals: heart-rate MAE of 1.994 bpm and respiratory-rate MAE of 0.197 bpm, with Huawei Watch GT 3 and a respiration belt as the respective references.

## Entrepreneurship & Applied AI

**Forget-Me-Not: Voice-Based Cognitive Screening** 2026–Present
*Co-founder & Algorithm Lead | Shouyi Technology*

- Lead the algorithm development of a telephone-based cognitive screening platform for older adults, connecting conversational agents, ASR/TTS, dialogue-state management, and speech/language feature extraction to cognitive-risk reports.
- Translate research into community and hospital pilots through real-time call pipelines, model evaluation, backend integration, and prototype deployment; work with clinical and community partners to support field testing.

## Selected Honors & Awards

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| 2026 | **Gold Award**, Alibaba Cloud C9 Hackathon |
| 2026 | **National Third Prize**, China Software Cup University Student Software Design Competition |
| 2026 | **National Third Prize, Maker Track**, 17th China College Students Service Outsourcing Innovation and Entrepreneurship Competition |
| 2026 | **Bronze Award**, NJUPT Challenge Cup Student Entrepreneurship Competition |
| 2026 | **Second-Class University Scholarship**, Nanjing University of Posts and Telecommunications |

## Technical Skills

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| **Programming** | Python, PyTorch, FastAPI, SQLite; backend APIs and asynchronous data pipelines |
| **Agent systems** | Tool calling, workflow orchestration, retrieval-augmented generation, memory, structured outputs, verifier-guided evaluation |
| **Speech & sensing** | ASR/TTS integration, speech/language features, cross-corpus evaluation, WiFi/mmWave sensor integration, multimodal prototyping |